

Quantum Tunneling Simulator: Formulas

Every formula used by the quantum tunneling simulator, in units $\hbar = m = 1$.

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1 Model

1.1 Schrödinger Equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi$$

The program uses units $\hbar = m = 1$, so it solves

$$i \frac{\partial \psi}{\partial t} = -\frac{1}{2} \frac{\partial^2 \psi}{\partial x^2} + V(x) \psi$$

Formal solution over one step: $\psi(t + \Delta t) = e^{-i\hat{H}\Delta t} \psi(t)$, with $\hat{H} = \hat{T} + \hat{V}$.

Used in: the whole simulation (`server.py`).

Related: [Rectangular Barrier Potential](#), [Split-Step Fourier Method](#), [Probability Density](#)

1.2 Rectangular Barrier Potential

$$V(x) = \begin{cases} V_0 & |x| < a/2 \\ 0 & \text{otherwise} \end{cases}$$

- V_0 : barrier height
- a : barrier width (the *width* box in the app)

Code: `V = np.where(np.abs(x) < width / 2, V0, 0.0)`

Related: [Schrödinger Equation](#), [Potential Step](#), [Transmission Below the Barrier](#)

1.3 Gaussian Wave Packet

Initial state:

$$\psi(x, 0) = (2\pi\sigma^2)^{-1/4} e^{-(x-x_0)^2/(4\sigma^2)} e^{ik_0x}$$

- $x_0 = -30$: starting position
- $\sigma = 2.5$: width (standard deviation of $|\psi|^2$)
- k_0 : central momentum (the k_0 box)

The code normalizes numerically so that $\int |\psi|^2 dx = 1$:

$$\psi \leftarrow \frac{\psi}{\sqrt{\sum_j |\psi_j|^2 \Delta x}}$$

The momentum spread is $\sigma_k = \frac{1}{2\sigma} = 0.2$. The peak density is $\frac{1}{\sqrt{2\pi}\sigma} \approx 0.1596$, which is why the y-axis starts at $1.3 \times 0.1596 \approx 0.2075$.

Code: `make_packet(k0)`

Related: [Packet Energy](#), [Wave Packet Spreading](#), [Probability Density](#)

1.4 Probability Density

$$\rho(x, t) = |\psi(x, t)|^2 = \psi^* \psi = (\text{Re } \psi)^2 + (\text{Im } \psi)^2$$

Total probability is conserved by the Schrödinger equation:

$$\int_{-\infty}^{\infty} \rho dx = 1$$

(In the simulation it only drops when the [Absorbing Boundary](#) removes the wave.)

Code: `density = np.abs(psi) ** 2` is the curve drawn on the plot.

Related: [Region Probabilities](#), [Gaussian Wave Packet](#)

1.5 Packet Energy

$$E = \frac{\hbar^2 k_0^2}{2m} = \frac{k_0^2}{2}$$

This decides the regime shown in the app:

- $E < V_0$: classically forbidden, the packet can only **tunnel**
- $E > V_0$: classically allowed (partial reflection still happens)

Because the packet has a spread of momenta, its mean energy is slightly higher:

$$\langle E \rangle = \frac{k_0^2}{2} + \frac{1}{8\sigma^2}$$

(only 0.02 for $\sigma = 2.5$).

Related: [Gaussian Wave Packet](#), [Transmission Below the Barrier](#), [Transmission Above the Barrier](#)

2 Numerical method

2.1 Split-Step Fourier Method

Strang splitting of $e^{-i(\hat{T}+\hat{V})\Delta t}$:

$$\psi(t + \Delta t) \approx e^{-iV\Delta t/2} \mathcal{F}^{-1} \left[e^{-ik^2\Delta t/2} \mathcal{F} \left[e^{-iV\Delta t/2} \psi(t) \right] \right]$$

- Unitary, so it conserves probability exactly (before absorption)
- Error is $O(\Delta t^2)$ globally
- $\Delta t = 0.004$, 20 steps per frame, 900 frames (72 time units)

Consecutive half-steps of the potential merge into a full step $e^{-iV\Delta t}$, which is why the code applies `full_abs` between FFTs and `half_abs` only at the end of a frame.

Code: `Simulation.step()`

Related: [Kinetic Step](#), [Potential Step](#), [Wavenumber Grid](#), [Absorbing Boundary](#)

2.2 Kinetic Step

In momentum space the kinetic energy is a simple multiplication:

$$\hat{\psi}(k) \rightarrow e^{-ik^2\Delta t/2} \hat{\psi}(k)$$

$$\text{since } T = \frac{\hbar^2 k^2}{2m} = \frac{k^2}{2}.$$

Code: `KIN = np.exp(-1j * k_grid**2 / 2 * DT)`

Related: [Split-Step Fourier Method](#), [Wavenumber Grid](#)

2.3 Potential Step

In position space the potential is a pointwise phase:

$$\psi(x) \rightarrow e^{-iV(x)\Delta t/2} \psi(x) \quad (\text{half step})$$

$$\psi(x) \rightarrow e^{-iV(x)\Delta t} \psi(x) \quad (\text{full step, two merged halves})$$

Code: `half_pot, and full_abs = exp(-1j * V * DT) * ABSORBER`

Related: [Split-Step Fourier Method](#), [Rectangular Barrier Potential](#), [Absorbing Boundary](#)

2.4 Wavenumber Grid

Grid spacing and the FFT's wavenumbers:

$$\Delta x = \frac{L}{N}, \quad k_j = \frac{2\pi}{N\Delta x} \times \begin{cases} j & 0 \leq j < N/2 \\ j - N & N/2 \leq j < N \end{cases}$$

With $L = 360$ and $N = 12288$: $\Delta x \approx 0.0293$ and $\Delta k = 2\pi/L \approx 0.0175$.

Code: `k_grid = 2 * np.pi * np.fft.fftfreq(N, d=dx)`

Related: [Kinetic Step](#), [Nyquist Limit](#)

2.5 Nyquist Limit

The grid can only represent wavenumbers up to

$$k_N = \frac{\pi}{\Delta x} \approx 107.2$$

A packet with larger k_0 aliases to a lower or negative wavenumber and appears to move the wrong way. The app allows only

$$k_0 < 0.75 k_N \approx 80.4$$

The 25% margin covers the packet's own momentum spread ($\sigma_k = 0.2$), whose upper tail would otherwise alias.

Code: K_NYQUIST, KO_SAFE_MAX

Related: [Wavenumber Grid](#), [Gaussian Wave Packet](#)

2.6 Absorbing Boundary

The FFT assumes a periodic box, so a wave leaving on the right would re-enter on the left. Near each end the wave is faded out smoothly:

$$\psi_j \leftarrow e^{-\eta(x_j) \Delta t} \psi_j, \quad \eta(x) = \eta_{\max} \left(\frac{|x| - x_s}{w} \right)^2 \quad (|x| > x_s), \quad 0 \text{ otherwise}$$

- $\eta_{\max} = 4$, layer width $w = 40$, start $x_s = L/2 - w = 140$
- Equivalent to an imaginary potential $V \rightarrow V - i\eta$
- The quadratic ramp turns on gradually, so there is almost no reflection (very slow packets, small k_0 , can reflect a little)

Code: ABSORBER = np.exp(-eta * DT)

Related: [Split-Step Fourier Method](#), [Potential Step](#), [Probability Density](#)

3 Analytic reference

3.1 Transmission Below the Barrier

Case $E < V_0$.

$$\kappa = \sqrt{2(V_0 - E)}$$

$$T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E(V_0 - E)} \right]^{-1}$$

For a thick barrier ($\kappa a \gg 1$) this reduces to

$$T \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa a}$$

so transmission falls off exponentially with width and with $\sqrt{V_0 - E}$.

This is the “single-energy analytic T ” in the app. It is a reference at the packet’s central energy, not the exact simulated result, because the packet contains a spread of energies.

Code: analytic_transmission() (first branch)

Related: [Packet Energy](#), [Transmission Above the Barrier](#), [Transmission at Barrier Height](#)

3.2 Transmission Above the Barrier

Case $E > V_0$.

$$k' = \sqrt{2(E - V_0)}$$

$$T = \left[1 + \frac{V_0^2 \sin^2(k'a)}{4E(E - V_0)} \right]^{-1}$$

Reflection is classically forbidden here but still happens, except at the resonances $k'a = n\pi$ where $T = 1$.

Code: `analytic_transmission()` (second branch)

Related: [Transmission Below the Barrier](#), [Transmission at Barrier Height](#)

3.3 Transmission at Barrier Height

Case $E = V_0$. The limit of the two formulas as $E \rightarrow V_0$:

$$T = \left[1 + \frac{V_0 a^2}{2} \right]^{-1}$$

(in general units: $1 + \frac{mV_0 a^2}{2\hbar^2}$).

Code: the `else` branch of `analytic_transmission()`

Related: [Transmission Below the Barrier](#), [Transmission Above the Barrier](#)

4 Results

4.1 Region Probabilities

The line is split into left of the barrier, inside it, and right of it:

$$P_L = \int_{x < -a/2} \rho dx, \quad P_B = \int_{|x| \leq a/2} \rho dx, \quad P_R = \int_{x > a/2} \rho dx$$

computed as sums $\sum_j \rho_j \Delta x$ over the grid.

After the packet has separated: $P_L \approx R$ (reflected) and $P_R \approx T$ (transmitted).

Code: `P_left`, `P_barrier`, `P_right` in `Simulation.observe()`

Related: [Probability Density](#), [Measurement Criterion](#)

4.2 Measurement Criterion

The simulated P_L and P_R are frozen the first time the packet has passed through the barrier:

1. The packet has *entered*: $P_B > 0.01$
2. Then it has *cleared*: $P_B < 10^{-3}$
3. And little probability was lost: $P_L + P_B + P_R > 0.9$

Step 1 matters because “ P_B is near zero” is trivially true before the packet arrives.

Code: `entered` and `measured` in `Simulation.observe()`

Related: [Region Probabilities](#), [Absorbing Boundary](#)

4.3 Wave Packet Spreading

A free Gaussian packet widens because its momentum components travel at different speeds:

$$\sigma(t) = \sigma_0 \sqrt{1 + \left(\frac{\hbar t}{2m\sigma_0^2} \right)^2}$$

With $\sigma_0 = 2.5$ the spreading time is $2\sigma_0^2 = 12.5$. Probability is conserved, so the peak height falls as $1/\sigma(t)$.

This explains the behavior on screen; the code does not compute it.

Related: [Gaussian Wave Packet](#), [Probability Density](#)